

Table 1: Sensitivity to the Number of CHSW Projection Directions  $K$ . The best results are highlighted in bold.

| Dataset    | $K$ | ACC $\uparrow$                    | $\Delta_{DP} \downarrow$          | $\Delta_{DP}^{\text{soft}} \downarrow$ |
|------------|-----|-----------------------------------|-----------------------------------|----------------------------------------|
| Pullback   | 60  | <b>0.819<math>\pm</math>0.019</b> | 0.293 $\pm$ 0.051                 | 0.257 $\pm$ 0.057                      |
|            | 120 | 0.758 $\pm$ 0.064                 | <b>0.187<math>\pm</math>0.122</b> | <b>0.176<math>\pm</math>0.096</b>      |
|            | 240 | 0.807 $\pm$ 0.040                 | 0.268 $\pm$ 0.088                 | 0.243 $\pm$ 0.075                      |
| Hyperbolic | 60  | 0.485 $\pm$ 0.055                 | <b>0.229<math>\pm</math>0.159</b> | 0.013 $\pm$ 0.009                      |
|            | 120 | <b>0.521<math>\pm</math>0.061</b> | 0.245 $\pm$ 0.158                 | <b>0.005<math>\pm</math>0.006</b>      |
|            | 240 | 0.513 $\pm$ 0.073                 | 0.281 $\pm$ 0.160                 | 0.008 $\pm$ 0.006                      |
| SPD        | 60  | 0.785 $\pm$ 0.048                 | 0.221 $\pm$ 0.130                 | 0.088 $\pm$ 0.057                      |
|            | 120 | <b>0.795<math>\pm</math>0.034</b> | <b>0.210<math>\pm</math>0.145</b> | 0.079 $\pm$ 0.068                      |
|            | 240 | 0.793 $\pm$ 0.034                 | 0.221 $\pm$ 0.137                 | <b>0.078<math>\pm</math>0.066</b>      |
| Adult      | 60  | <b>0.843<math>\pm</math>0.005</b> | 0.115 $\pm$ 0.008                 | 0.098 $\pm$ 0.017                      |
|            | 120 | 0.840 $\pm$ 0.007                 | 0.098 $\pm$ 0.034                 | 0.082 $\pm$ 0.033                      |
|            | 240 | 0.823 $\pm$ 0.040                 | <b>0.081<math>\pm</math>0.054</b> | <b>0.067<math>\pm</math>0.048</b>      |
| LFWA       | 60  | 0.902 $\pm$ 0.011                 | 0.059 $\pm$ 0.010                 | 0.058 $\pm$ 0.009                      |
|            | 120 | <b>0.905<math>\pm</math>0.004</b> | <b>0.048<math>\pm</math>0.012</b> | <b>0.046<math>\pm</math>0.010</b>      |
|            | 240 | 0.902 $\pm$ 0.005                 | 0.057 $\pm$ 0.009                 | 0.051 $\pm$ 0.009                      |

Table 2: Runtime (ms/epoch) and Memory (MB) comparisons. Runtime is averaged per epoch over 10 epochs; for WB (post-processing), we report its one-time cost. Memory denotes peak GPU usage during training.

| Method           | Pullback |        | Hyperbolic |        | SPD     |        | Adult   |        | LFWA    |        |
|------------------|----------|--------|------------|--------|---------|--------|---------|--------|---------|--------|
|                  | Runtime  | Memory | Runtime    | Memory | Runtime | Memory | Runtime | Memory | Runtime | Memory |
| M <sub>AVG</sub> | 41.9     | 0.8    | 240.0      | 0.7    | 17.1    | 18.6   | 765.3   | 1.6    | 1680.3  | 188.6  |
| M <sub>MMD</sub> | 36.3     | 2.3    | 256.2      | 2.2    | 21.9    | 20.2   | 990.6   | 3.2    | 1673.7  | 188.6  |
| M <sub>SNK</sub> | 420.7    | 2.3    | 3098.0     | 2.3    | 255.9   | 20.2   | 18115.7 | 22.9   | 4058.8  | 188.6  |
| WB               | 25.6     | 0.8    | 160.8      | 0.7    | 18.2    | 18.6   | 483.7   | 1.6    | 1828.7  | 358.5  |
| SW               | 43.0     | 2.2    | 297.9      | 2.2    | 26.2    | 19.2   | 920.3   | 3.1    | 2187.1  | 358.5  |
| Ours             | 48.9     | 2.3    | 321.9      | 6.9    | 36.0    | 19.3   | 925.5   | 3.1    | 2095.7  | 188.6  |

Table 3: Sensitivity to the Choice of Origin  $o$ . The best results are highlighted in bold.

| Dataset    | Origin                 | Description                               | ACC $\uparrow$                    | $\Delta_{DP} \downarrow$          | $\Delta_{DP}^{\text{soft}} \downarrow$ |
|------------|------------------------|-------------------------------------------|-----------------------------------|-----------------------------------|----------------------------------------|
| Hyperbolic | center ( $\ o\  = 0$ ) | canonical origin (Poincaré ball center)   | <b>0.521<math>\pm</math>0.061</b> | 0.245 $\pm$ 0.158                 | <b>0.005<math>\pm</math>0.006</b>      |
|            | $\ o\  = 0.15$         | random point with small displacement      | 0.502 $\pm$ 0.065                 | 0.236 $\pm$ 0.153                 | 0.009 $\pm$ 0.008                      |
|            | $\ o\  = 0.30$         | random point with larger displacement     | 0.477 $\pm$ 0.052                 | <b>0.226<math>\pm</math>0.160</b> | 0.011 $\pm$ 0.008                      |
|            | data_mean              | heuristic: Fréchet mean of all samples    | 0.474 $\pm$ 0.052                 | 0.245 $\pm$ 0.142                 | 0.010 $\pm$ 0.008                      |
|            | group_barycenter       | adaptive: fairness-aware group barycenter | 0.504 $\pm$ 0.064                 | 0.234 $\pm$ 0.152                 | 0.009 $\pm$ 0.008                      |
| SPD        | identity ( $I$ )       | canonical origin (identity matrix)        | <b>0.795<math>\pm</math>0.034</b> | 0.210 $\pm$ 0.145                 | 0.079 $\pm$ 0.068                      |
|            | random1                | random SPD matrix (seed=42)               | 0.778 $\pm$ 0.044                 | 0.188 $\pm$ 0.197                 | 0.081 $\pm$ 0.087                      |
|            | random2                | random SPD matrix (seed=123)              | 0.793 $\pm$ 0.037                 | <b>0.179<math>\pm</math>0.159</b> | <b>0.073<math>\pm</math>0.076</b>      |
|            | data_mean              | heuristic: Fréchet mean of all samples    | 0.789 $\pm$ 0.050                 | 0.249 $\pm$ 0.167                 | 0.120 $\pm$ 0.077                      |
|            | group_mean             | adaptive: group-wise barycenter           | 0.790 $\pm$ 0.047                 | 0.263 $\pm$ 0.169                 | 0.125 $\pm$ 0.082                      |

Table 4: Comparison with Adversarial FRL Baselines on Adult and LFWA. The best results are highlighted in bold.

| Method | Adult                             |                                   |                                        | LFWA                              |                                   |                                        |
|--------|-----------------------------------|-----------------------------------|----------------------------------------|-----------------------------------|-----------------------------------|----------------------------------------|
|        | ACC $\uparrow$                    | $\Delta_{DP} \downarrow$          | $\Delta_{DP}^{\text{soft}} \downarrow$ | ACC $\uparrow$                    | $\Delta_{DP} \downarrow$          | $\Delta_{DP}^{\text{soft}} \downarrow$ |
| CHSW   | <b>0.840<math>\pm</math>0.007</b> | <b>0.098<math>\pm</math>0.034</b> | <b>0.082<math>\pm</math>0.033</b>      | <b>0.905<math>\pm</math>0.004</b> | <b>0.048<math>\pm</math>0.012</b> | <b>0.046<math>\pm</math>0.010</b>      |
| LAFTR  | 0.835 $\pm$ 0.003                 | 0.168 $\pm$ 0.009                 | 0.176 $\pm$ 0.003                      | 0.893 $\pm$ 0.004                 | 0.074 $\pm$ 0.006                 | 0.068 $\pm$ 0.004                      |
| HAM    | 0.839 $\pm$ 0.002                 | 0.163 $\pm$ 0.010                 | 0.180 $\pm$ 0.007                      | 0.901 $\pm$ 0.004                 | 0.062 $\pm$ 0.006                 | 0.057 $\pm$ 0.008                      |